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A Dual-Target Explainable AI System for Early Prediction of Dengue Shock Risk and Platelet Nadir Forecasts

ABSTRACT

Current tools for severe dengue often act like "black boxes" and provide only basic binary risk alerts, failing to offer quantitative forecasts for clinical resource planning. To solve this, our Dual-Target Early Warning System uses advanced Machine Learning and Explainable AI (SHAP) to transform early admission data into transparent, actionable insights. Our system simultaneously predicts severe shock risk and forecasts the exact platelet nadir, giving doctors a simple dashboard to make quick, life-saving decisions right at the patient's bedside.

OBJECTIVE

- To develop an interactive, real-time dengue monitoring dashboard.
- To predict the early warning risk of severe dengue shock using ensemble machine learning.
- To forecast the quantitative platelet nadir for blood bank resource planning.
- To ensure clinical trust by integrating feature-level Explainable AI (SHAP).

METHODOLOGY

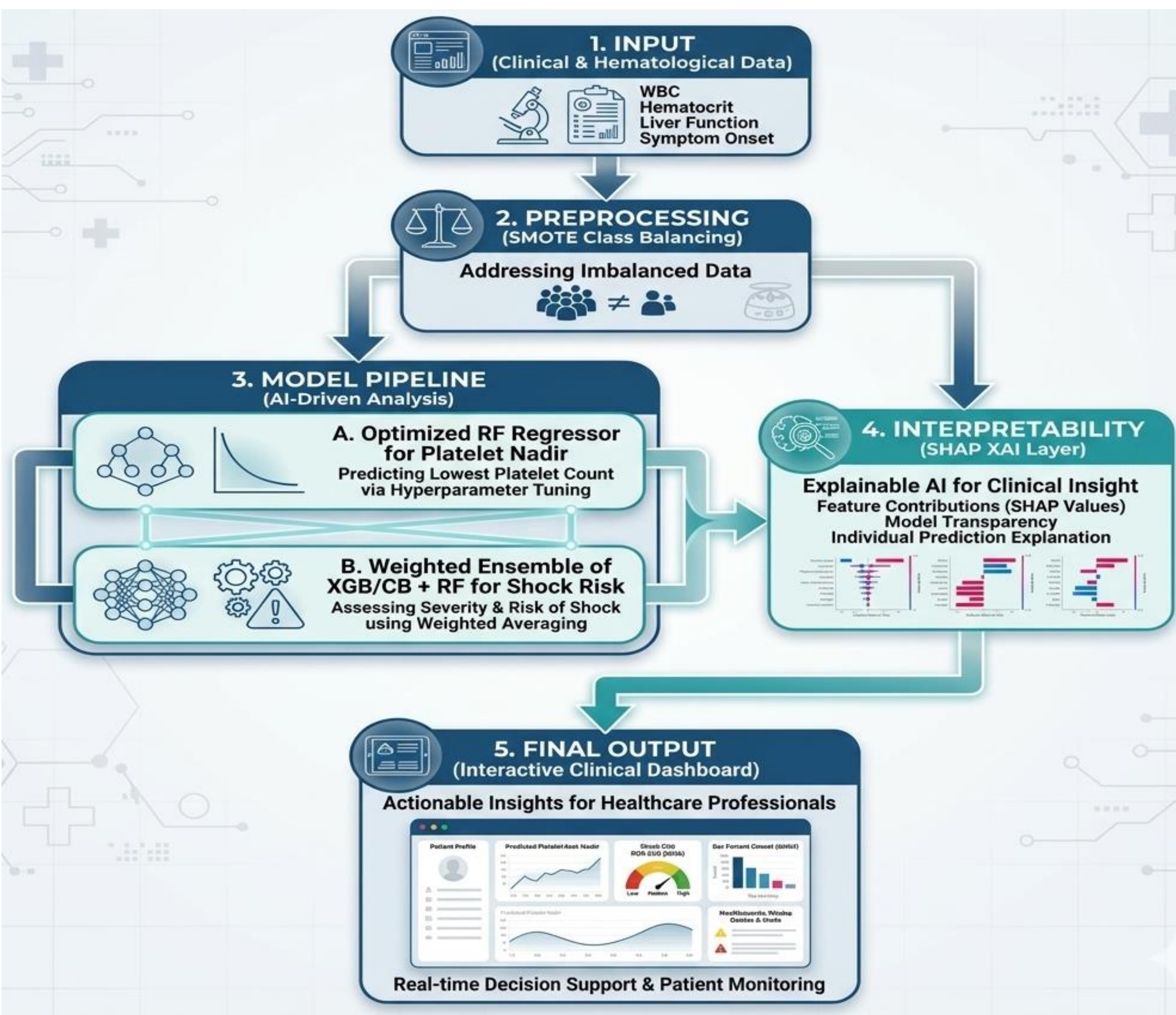


Figure No.01: Methodology

RESULT

Transforming Static Clinical Markers into Actionable, Real-Time Bedside foresight.

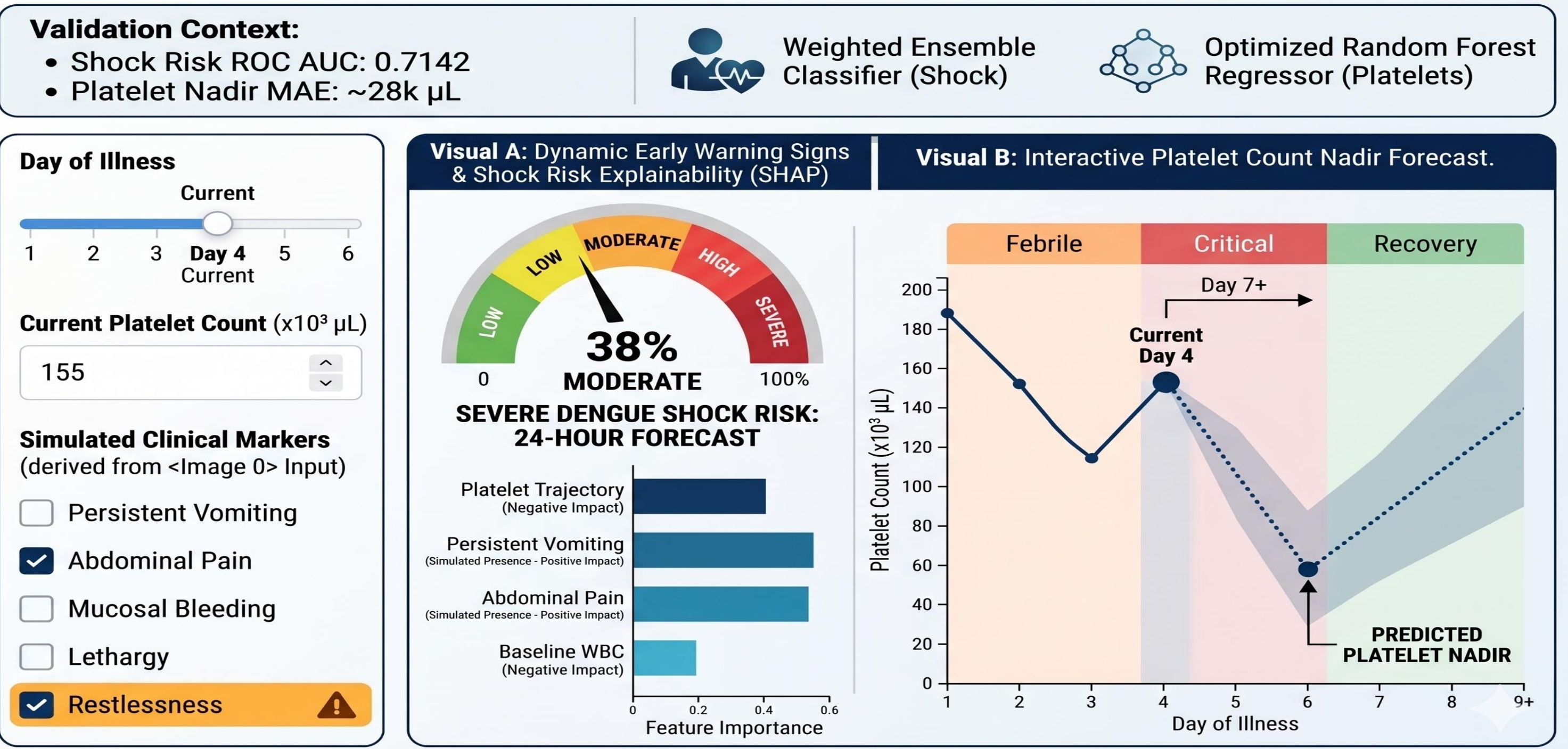


Figure No.03: interactive results dashboard

CONCLUSION

Deployed as an interactive clinical dashboard, this dual-target system transforms early static markers into actionable prognostic foresight. By providing both qualitative risk alerts and quantitative platelet forecasts backed by Explainable AI, this framework serves as a vital blueprint for intelligent, real-time triage in resource-constrained environments. Future iterations aim to integrate diverse longitudinal datasets across all age groups.

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